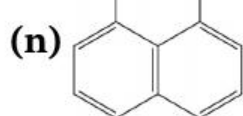
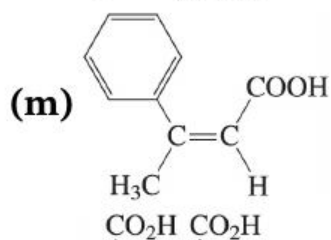
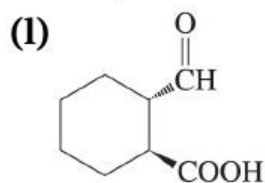
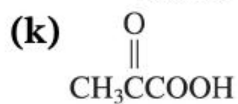
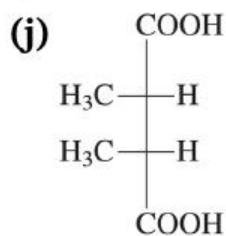
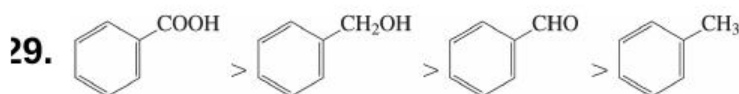


Solutions to Problems

27. (a) 2-Chloro-4-methylpentanoic acid;
 (b) 2-ethyl-3-butenic acid
 (c) *E*-2-bromo-3,4-dimethyl-2-pentenoic acid;
 (d) cyclopentylacetic acid;
 (e) *trans*-2-hydroxycyclohexanecarboxylic acid;
 (f) *E*-2-chlorobutenedioic acid;
 (g) 2,4-dihydroxy-6-methylbenzoic acid;
 (h) 1,2-benzenedicarboxylic acid;
 (i) $\text{H}_2\text{NCH}_2\text{CH}_2\text{CH}_2\text{COOH}$ $\text{H}_2\text{NCH}_2\text{CH}_2\text{CH}_2\text{COOH}$



28. (a) 3-Chloro-4-hydroxy-2-butanone;
 (b) Z-2-ethyl-3-formyl-2-pentenoic acid;
 (c) 5-hydroxycyclopent-1-enecarboxylic acid or 2-hydroxy-1-cyclopentenecarboxylic acid;
 (d) 3-acetyl-5-nitrobenzenecarboxylic acid or 3-acetyl-5-nitrobenzoic acid.

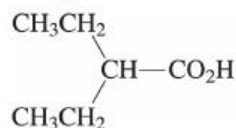


The order is the same for both boiling point and solubility in water. The acid has the most hydrogen-bonding capability and will have the highest boiling point (249°C)(249°C) because of its hydrogen-bonded dimer formation. The alcohol, which can also hydrogen bond, is next (205°C),(205°C), the polar aldehyde third (178°C),(178°C), and the nearly nonpolar hydrocarbon last (115°C).(115°C). Solubilities follow similar considerations, except that the acid and alcohol are quite similar in water solubility because they can both hydrogen bond with H₂O.H₂O.

30. (a) Order is as written;
 (b) order is reverse of that given;
 (c) $\text{CH}_3\text{CH}_2\text{CHClCO}_2\text{H} > \text{CH}_3\text{CHClCH}_2\text{CO}_2\text{H} > \text{ClCH}_2\text{CH}_2\text{CH}_2\text{CO}_2\text{H}$
 $\text{CH}_3\text{CH}_2\text{CHClCO}_2\text{H} > \text{CH}_3\text{CHClCH}_2\text{CO}_2\text{H} > \text{ClCH}_2\text{CH}_2\text{CH}_2\text{CO}_2\text{H};$
 (d) order is as written;
 (e) 2,4-dinitro>4-nitro>unsubstituted>4-methoxybenzoic_(2,4-dinitro > 4-nitro > unsubstituted > 4-methoxybenzoic)
31. Begin by scanning the given information to look for data that stand out for being particularly diagnostic. The mass information lends itself only to trial-and-error narrowing down of molecular formulas. Pass on that for starters. The IR,IR, on the other hand, is unambiguously diagnostic for the CO₂HCO₂H functional group of a

carboxylic acid. The $^1\text{H NMR}$ confirms, with a signal near $\delta=12\text{ ppm}$. Three other $^1\text{H NMR}$ signals, and four $^{13}\text{C NMR}$ signals total, suggest a four-carbon acid. The simplest, and a reasonable first guess, is butanoic acid, $\text{CH}_3\text{CH}_2\text{CH}_2\text{CO}_2\text{H}$. But its mass is only 88. We are missing 28 mass units, suggesting we still have work to do.

The $^1\text{H NMR}$ splitting patterns confirm: *two* equivalent CH_3 groups appear as a 6 H triplet near $\delta=0.9\text{ ppm}$, which can mean only that the molecule contains two equivalent CH_3CH_2 groups. The two CH_2 groups give the 4 H signal at $\delta=1.6\text{ ppm}$. Finally, a 1 H quintet near $\delta=2.4\text{ ppm}$ must be a CH attached to both CH_2 groups, and by process of elimination, the COOH as well, giving as our solution the compound



A check of the mass: $\text{C}_6\text{H}_{12}\text{O}_2$ gives $72+12+32=116$.
 $72 + 12 + 32 = 116$.

32. (a) $H_{\text{sat}}=14+2=16$; degree of unsaturation $= (16-12)/2 = 2$ π bonds or rings are present.

$$H_{\text{sat}} = 14 + 2 = 16; \text{ degree of unsaturation} = \frac{(16 - 12)}{2} = 2 \pi \text{ bonds or rings are present.}$$

Compare Figure 19-3. This is a carboxylic acid

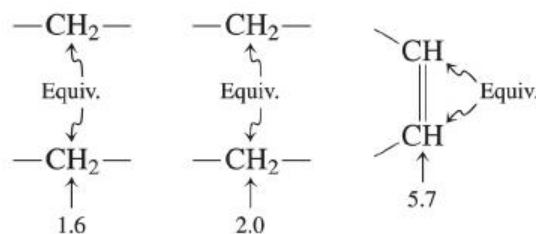
$$(\tilde{\nu}=1704 \text{ and } 3040 \text{ cm}^{-1}). (\tilde{\nu} = 1704 \text{ and } 3040 \text{ cm}^{-1}).$$

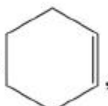
- (b) For B, $H_{\text{sat}}=12+2=14$; degree of unsaturation $= (14-10)/2 = 2$ π bonds or rings.

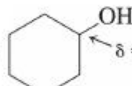
$$\text{B, } H_{\text{sat}} = 12 + 2 = 14; \text{ degree of unsaturation} = \frac{(14 - 10)}{2} = 2 \pi \text{ bonds or rings.}$$

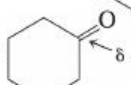
From the $^{13}\text{C NMR}$ (three signals), the molecule would seem to have twofold symmetry, with two sets of equivalent pairs of alkyl carbons and a pair of equivalent alkene carbons. The $^1\text{H NMR}$ shows two equal-area upfield signals (4 H

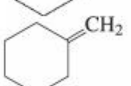
each) and a 2 H² H alkene signal. So the pieces seem to be

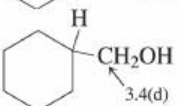


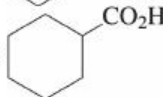
which can be simply put together to give , cyclohexene!

Then **C** =  $\delta = 69.5(^{13}\text{C})$ via oxymercuration–demercuration;

D =  $\delta = 208.5(^{13}\text{C})$ ($\tilde{\nu}_{\text{C=O}} = 1715 \text{ cm}^{-1}$);

E =  ($\tilde{\nu}_{\text{C=C}} = 1649 \text{ cm}^{-1}$ and $\tilde{\nu}_{\text{C=CH}_2} = 888 \text{ cm}^{-1}$) via Wittig reaction;

F =  $\delta = 3.4(\text{d})$ ($\tilde{\nu}_{\text{O-H}} = 3328 \text{ cm}^{-1}$) via hydroboration–oxidation;

A = 

(c) For G, $H_{\text{sat}} = 16 + 2 = 18$; degree of unsaturation = $(18 - 14)/2 = 2$ π bonds or rings.

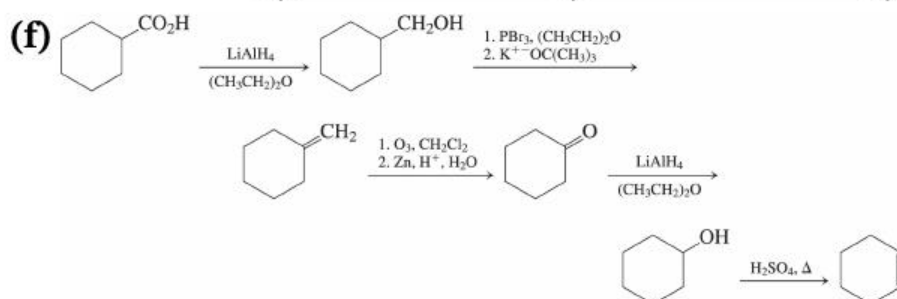
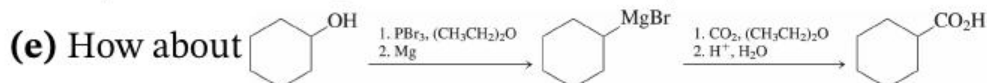
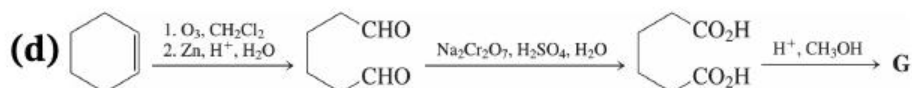
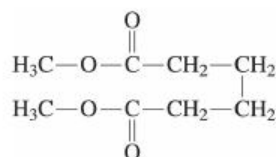
G, $H_{\text{sat}} = 16 + 2 = 18$; degree of unsaturation = $\frac{(18 - 14)}{2} = 2$ π bonds or rings.

IR: $\tilde{\nu} = 1742 \text{ cm}^{-1}$ IR: $\tilde{\nu} = 1742 \text{ cm}^{-1}$ is C=O , $\text{C}=\text{O}$, very possibly an ester because of the high value and the large number of oxygens in the formula.

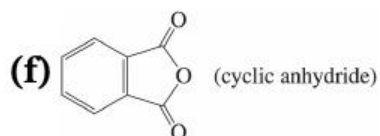
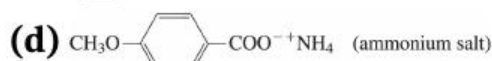
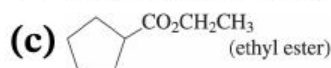
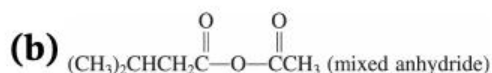
NMR: Only three signals, with integrations of 4, 4, and 6 H. The molecule must be symmetrical, with pieces such as

$2 \text{ H}_3\text{C}-\text{O}-$ ($\delta = 3.7$), $2 \text{ H}_3\text{C}-\text{O}-$ ($\delta = 3.7$), $2 -\text{CH}_2-\text{C}(=\text{O})-$ ($\delta = 2.4$), ($\delta = 2.4$), and two more equivalent $-\text{CH}_2-$'s ($\delta = 1.7$).

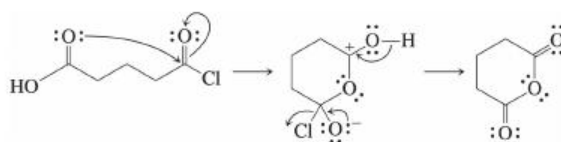
$-\text{CH}_2-$'s ($\delta = 1.7$). The splitting between the upfield signals suggests that the sets of CH_2 's are connected, so a reasonable answer is



33. (a) $(\text{CH}_3)_2\text{CH}_2\text{CH}_2\text{COCl}(\text{CH}_3)_2\text{CH}_2\text{CH}_2\text{COCl}$ (acyl chloride)



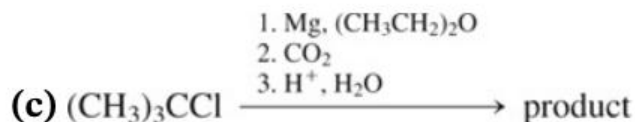
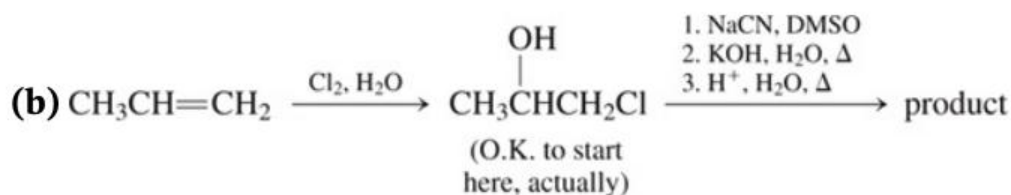
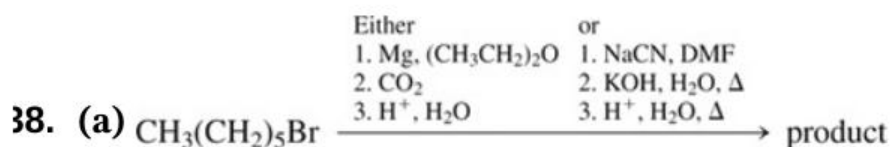
34. Recall ([Section 19-8](#)) that carboxylic acids may react with acyl halides to produce anhydrides. Upon conversion of one carboxylic acid function of a 1,4- or 1,5-diacid to a halide, the other acid function may react with it intramolecularly to form the cyclic anhydride. For example:



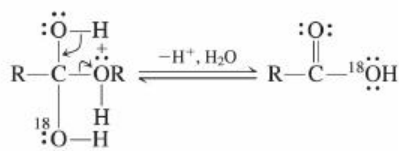
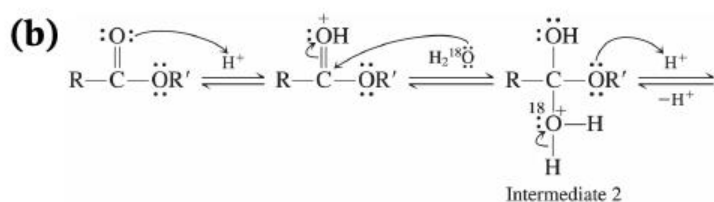
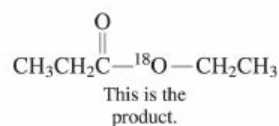
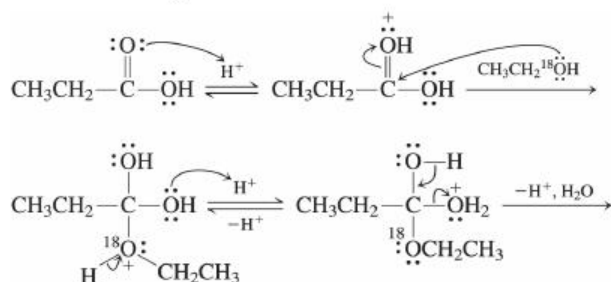
Are you surprised that I used the oxygen of the carbonyl group to form the ring instead of the OO of the carboxy —OH—OH group? Which oxygen is more basic (and therefore more nucleophilic)? Check out Section 19-4 in the textbook.

- 35.** (a) any of numerous oxidants such as CrO_3 , CrO_3 , normally in aqueous solvent;
 (b) $\text{H}+\text{H}^+$ or $-\text{OH}, \text{H}_2\text{O}, ^-\text{OH}, \text{H}_2\text{O}$, heat;
 (c) 1. Mg, Mg , ether, 2. CO_2, CO_2 , 3. $\text{H}+, \text{H}_2\text{O}, \text{H}^+, \text{H}_2\text{O}$ or 1. $-\text{CN}, ^-\text{CN}$, 2. $\text{H}+\text{H}^+$ or $-\text{OH}, \text{H}_2\text{O}, ^-\text{OH}, \text{H}_2\text{O}$, heat;
 (d) $\text{CrO}_3, \text{H}_2\text{SO}_4, \text{H}_2\text{O}; \text{CrO}_3, \text{H}_2\text{SO}_4, \text{H}_2\text{O}$;
 (e) $\text{H}+\text{H}^+$ or $-\text{OH}, \text{H}_2\text{O}, ^-\text{OH}, \text{H}_2\text{O}$, heat.
- 36.** (a) $\text{LiAlH}_4; \text{LiAlH}_4$;
 (b) hexanoyl chloride (see part c);
 (c) $\text{SOCl}_2; \text{SOCl}_2$;
 (d) $\text{Br}_2, \text{P}; \text{Br}_2, \text{P}$;
 (e) $\text{CH}_3\text{CH}_2\text{OH}, \text{H}+; \text{CH}_3\text{CH}_2\text{OH}, \text{H}^+$;
 (f) 1. $\text{SOCl}_2, \text{SOCl}_2$, 2. Excess NH_3, NH_3 .

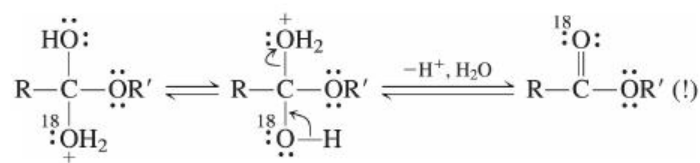
37. (a) $\text{Na}_2\text{Cr}_2\text{O}_7, \text{H}_2\text{O}, \text{H}_2\text{SO}_4; \text{Na}_2\text{Cr}_2\text{O}_7, \text{H}_2\text{O}, \text{H}_2\text{SO}_4;$
- (b) 1. $\text{NaCN}, \text{H}_2\text{O}, \text{H}_2\text{SO}_4, \text{NaCN}, \text{H}_2\text{O}, \text{H}_2\text{SO}_4$, 2. $\text{H}^+, \text{H}_2\text{O}, \Delta$;
 $\text{H}^+, \text{H}_2\text{O}, \Delta$;
- (c) 1. $\text{Mg}, \text{Mg}, (\text{CH}_3\text{CH}_2)_2\text{O}, (\text{CH}_3\text{CH}_2)_2\text{O}$, 2. CO_2, CO_2 , 3. $\text{H}^+, \text{H}_2\text{O}$;
 $\text{H}^+, \text{H}_2\text{O}$;
- (d) 1. $\text{NaCN}, \text{DMSO}, \text{NaCN}, \text{DMSO}$, 2. $\text{KOH}, \text{H}_2\text{O}, \Delta, \text{KOH}, \text{H}_2\text{O}, \Delta$, 3.
 $\text{H}^+, \text{H}_2\text{O}, \Delta; \text{H}^+, \text{H}_2\text{O}, \Delta$;
- (e) 1. $\text{SOCl}_2, \text{SOCl}_2$ (makes acyl chloride), 2. Add one more mole of
starting acid, $\Delta; \Delta$;
- (f) $(\text{CH}_3)_2\text{CHOH}, \text{H}^+; (\text{CH}_3)_2\text{CHOH}, \text{H}^+$;
- (g) $\text{CH}_3\text{COOH}, \Delta; \text{CH}_3\text{COOH}, \Delta$



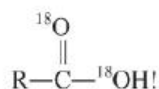
39. (a) Acid catalysis is understood for esterification. So,

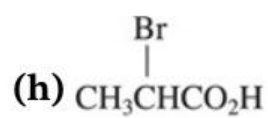
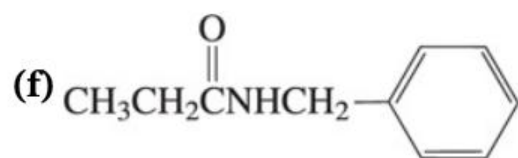
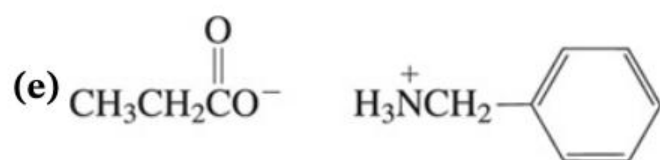
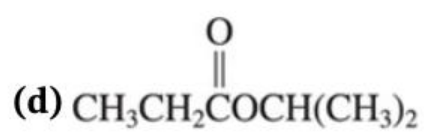
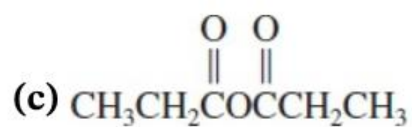
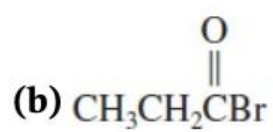
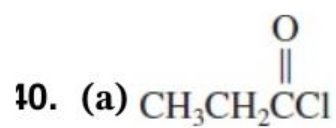


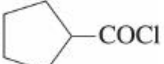
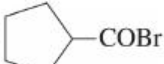
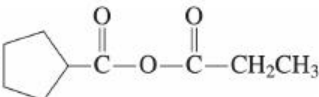
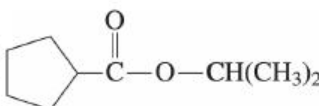
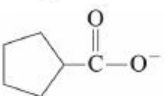
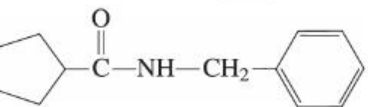
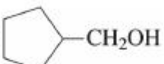
Alternatively, intermediate 2 could protonate on the unlabeled OH group instead of the OR'OR' group:



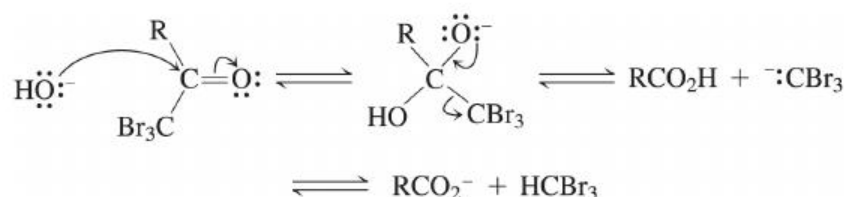
Ester containing ^{18}O in the carbonyl oxygen. Now if you follow the hydrolysis mechanism written above, the product will be





11. (a)  (b) 
 (c)  (d) 
 (e)   (a salt)
 (f) 
 (g) 
 (h) 

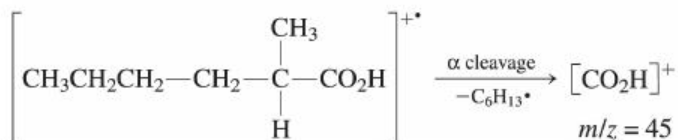
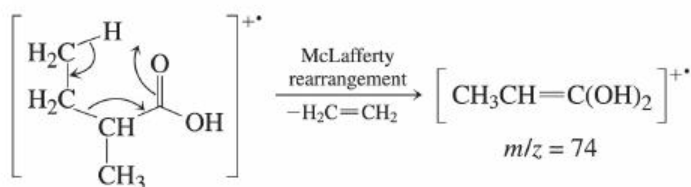
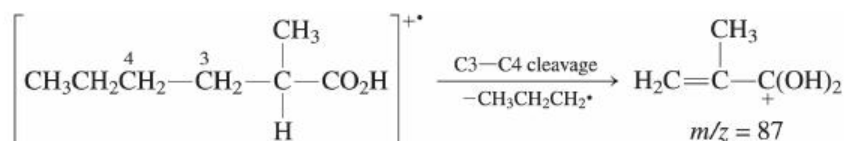
12. The only sensible way to start is by addition of the main nucleophile present (hydroxide) to the carbonyl carbon of the ketone. The resulting tetrahedral intermediate can be taken in the direction of the product by eliminating a novel leaving group, a tribromomethyl carbanion:



This carbanion is capable of leaving because its negative charge is stabilized by the combined inductive effects of the three halogen atoms. It is still a moderately strong base with a $\text{p}K_{\text{a}}^{\text{p}K_{\text{a}}}$ in the mid-teens. Proton transfer between this anion and the carboxylic acid gives the final products of the reaction, the salt of the acid and the haloform, bromoform

(CHBr₃).(CHBr₃). The same process occurs with trichloro- and triiodo-substituted methyl ketones, to give chloroform and iodoform, respectively.

13. Here are the processes that give rise to the labeled peaks.



14. 1. LiAlH₄, (CH₃CH₂)₂O; LiAlH₄, (CH₃CH₂)₂O (makes 1-pentanol); 2. KBr, H₂SO₄, Δ; KBr, H₂SO₄, Δ; 3. KCN, DMSO; KCN, DMSO (makes hexanenitrile); 4. KOH, H₂O, Δ; KOH, H₂O, Δ; 5. H⁺, H₂O, Δ; H⁺, H₂O, Δ

15. (a) SOCl₂; SOCl₂;

(b) H⁺, CH₃OH; H⁺, CH₃OH;

(c) H⁺, H⁺, 2-butanol;

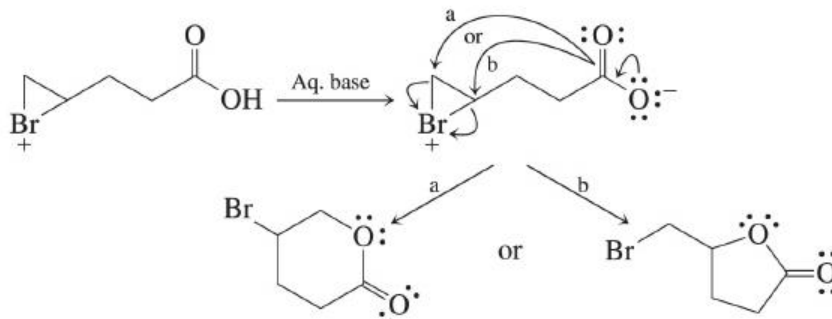
(d) acyl chloride from (a);

(e) CH₃NH₂; CH₃NH₂ (via the ammonium salt), Δ; Δ;

(f) LiAlH₄, (CH₃CH₂)₂O; LiAlH₄, (CH₃CH₂)₂O;

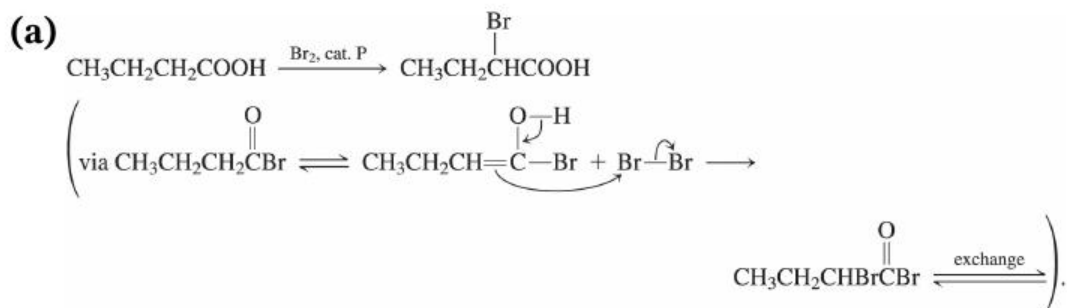
(g) Br₂; Br₂, cat. PP

16. As usual, think mechanistically. The first intermediate should be the cyclic bromonium ion:

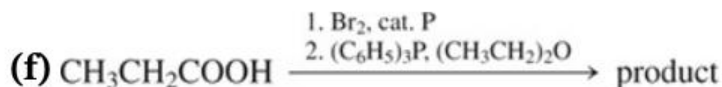
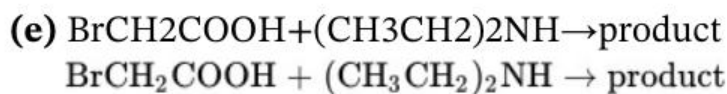
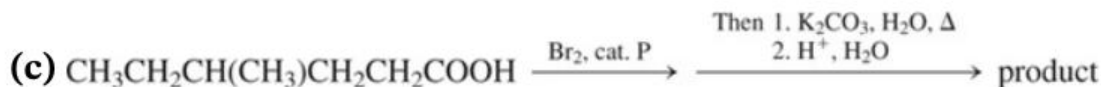
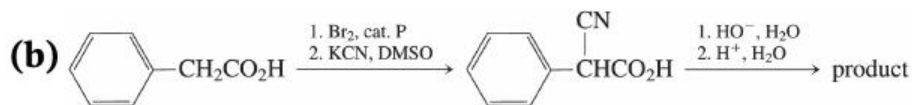
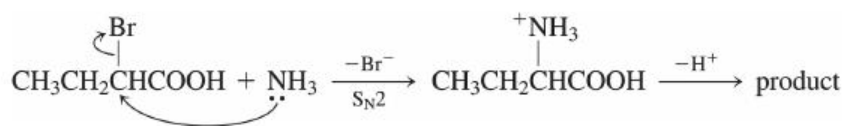


The bromolactones shown are formed by intramolecular attack by a carboxylate oxygen on one of the carbon atoms of a cyclic bromonium ion, the initial product of Br_2 addition to the double bond ([Section 12-6](#)). Which pathway, a or b, should be favored? The five-membered ring should form preferentially, for two reasons. First, attack by a nucleophile on a brominium ion normally takes place at the more highly substituted carbon atom (which is more positively polarized — see [Section 12-6](#) again). Second, five-membered rings form faster than do sixes ([Section 9-6](#)), compensating for the slight thermodynamic edge that six-membered rings have.

17.

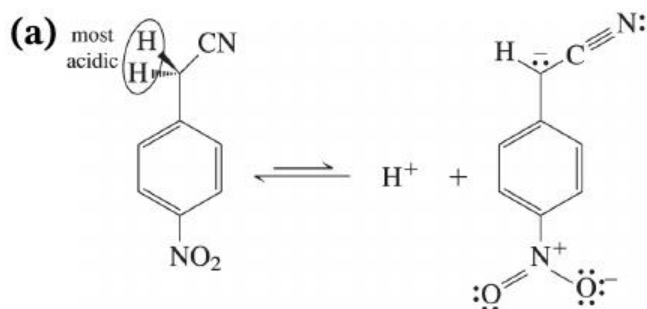


Then

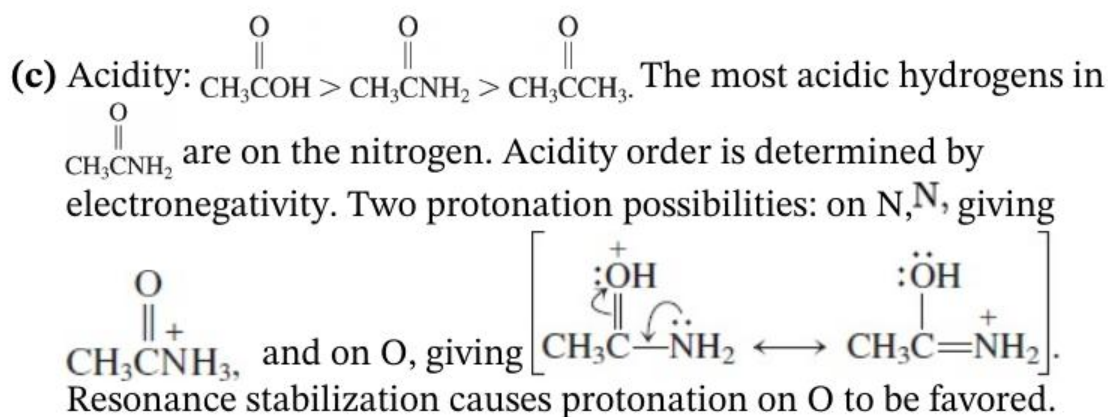
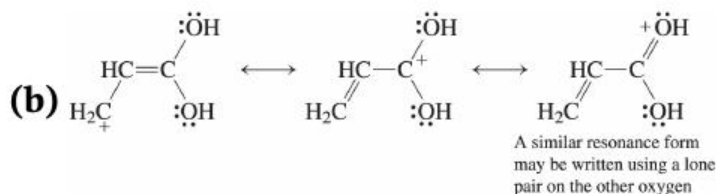


18. No tricks here! The mechanism is almost exactly as shown in Section 19-12, with only minor differences. Because the method starts with an acyl halide, step 1 is unnecessary. Step 2 (enolization) occurs as written. The third step uses Cl_2 , Cl_2 , generated in low concentration from NCS , NCS , or Br_2 , Br_2 , formed in the same way from NBS , NBS , or I_2 . I_2 . The fourth step in the mechanism doesn't apply because only acyl halides are present, not carboxylic acids.

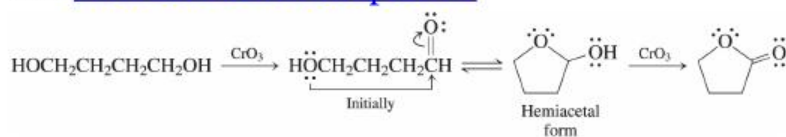
19.



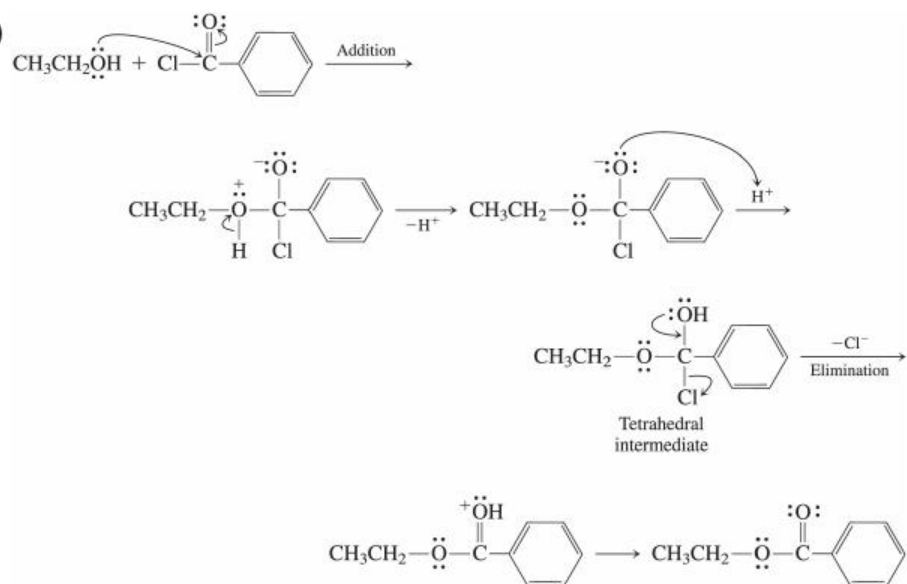
The conjugate base is stabilized by resonance. The lone pair (and negative charge) on the indicated carbon atom is delocalized both into the CNCN group above as well as the benzene ring and para nitro group below. Write them out!



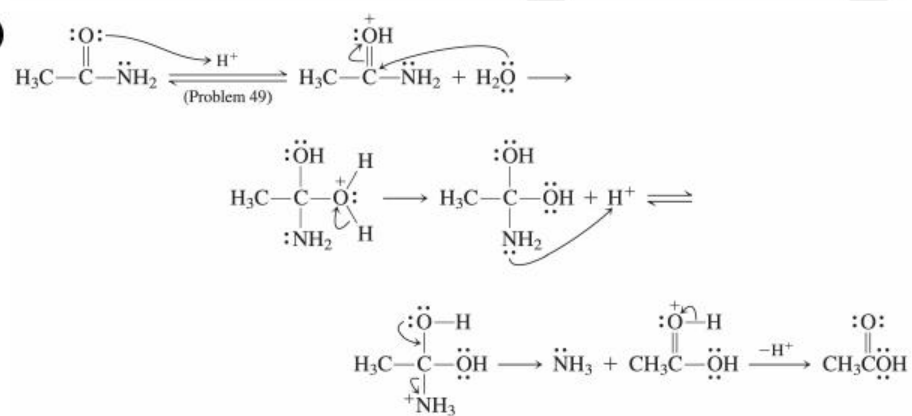
50. See [Problem 42 in Chapter 17](#).



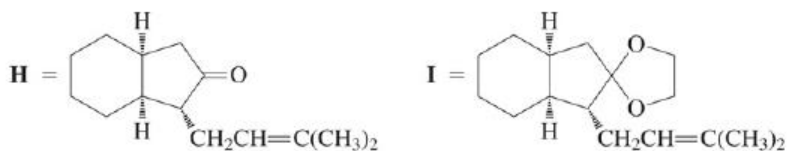
51. (a)



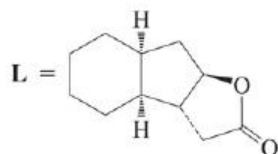
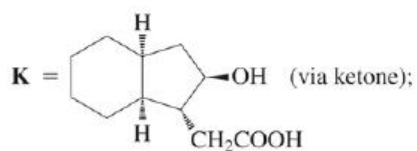
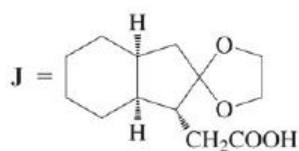
(b)



52.



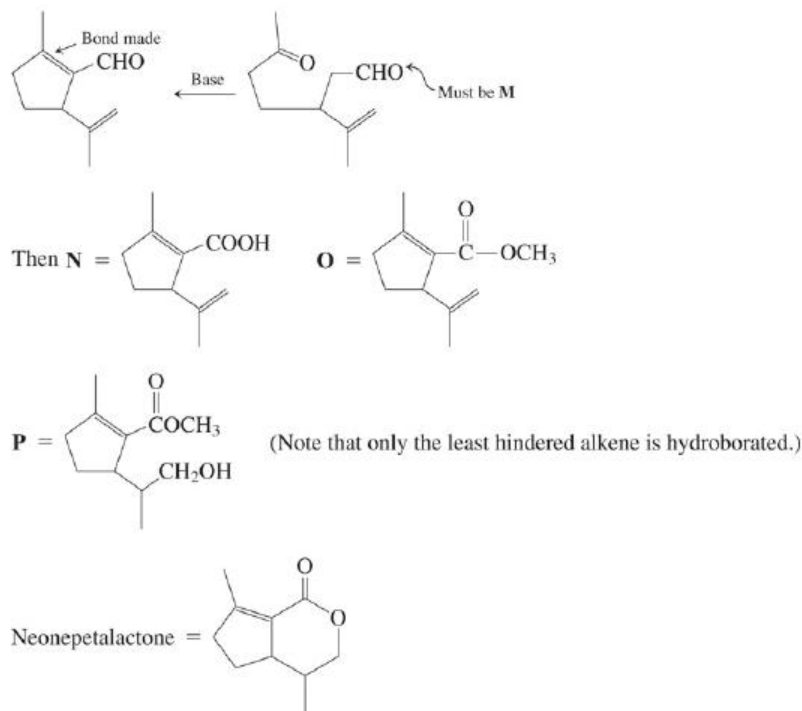
(Alkylation on less hindered bottom face)



A lactone ($\tilde{\nu}_{\text{C}=\text{O}} = 1770 \text{ cm}^{-1}$)

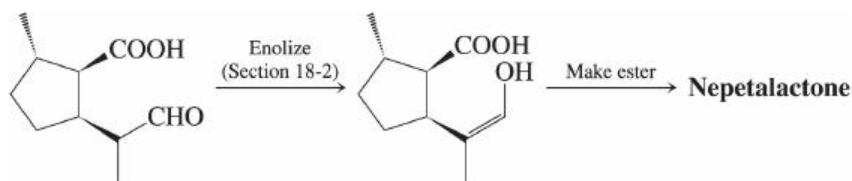
53. (a) Haloalkanes are generally not very soluble in water (too much polarity difference). The reaction mixture is heterogeneous, which prevents good mixing of reactants. Water also forms hydrogen bonds to the nucleophile, which doesn't help.
- (b) Acetic acid is a better solvent for the haloalkane, so the system is homogeneous, allowing for better mixing of reacting molecules.
- (c) Sodium dodecanoate is a soap, and it dissolves in water to make *micelles*. The less polar interior regions of the micelles form good solvents for molecules of low polarity like haloalkanes. The iodobutane dissolves **in the micelles** and is therefore in close proximity to the nucleophilic carboxylate groups, allowing the $\text{S}_{\text{N}}2$ reaction to proceed.

54. (a) Work forward and backward from the given structure. The first reaction looks like an aldol condensation.

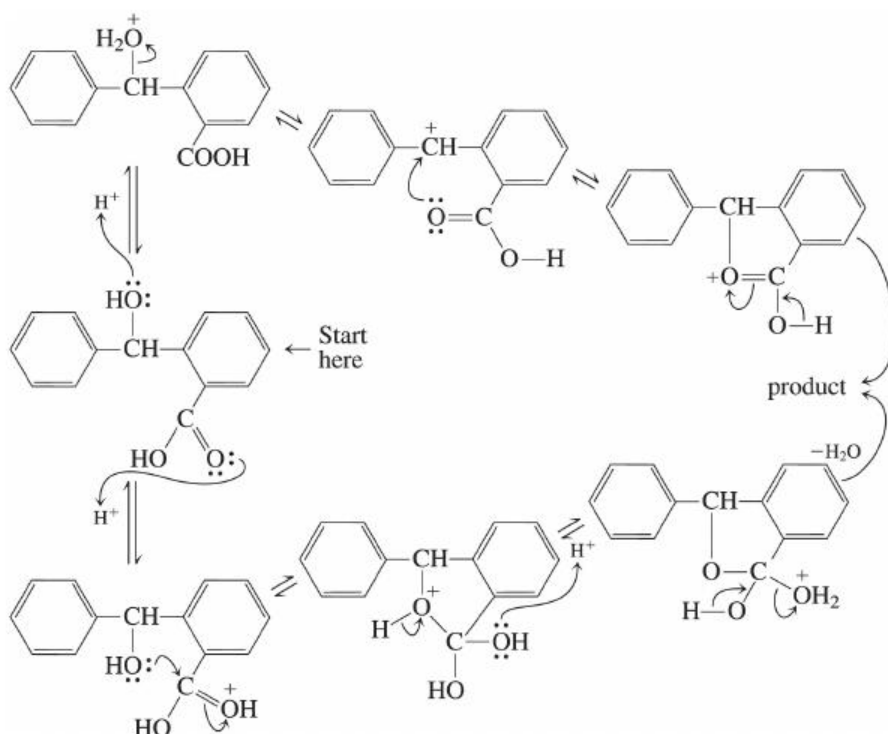


- (b) **Hint:** See the C=C double bond in the six-membered ether ring? What can you do to the aldehyde to create that double bond?

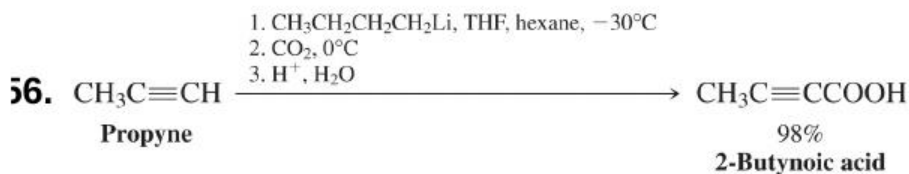
Answer: Convert it to its enol. Therefore:



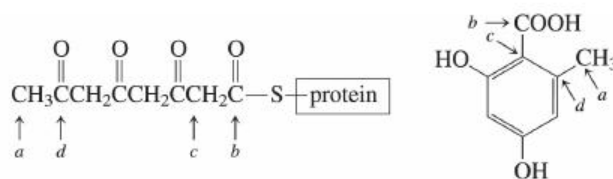
55.



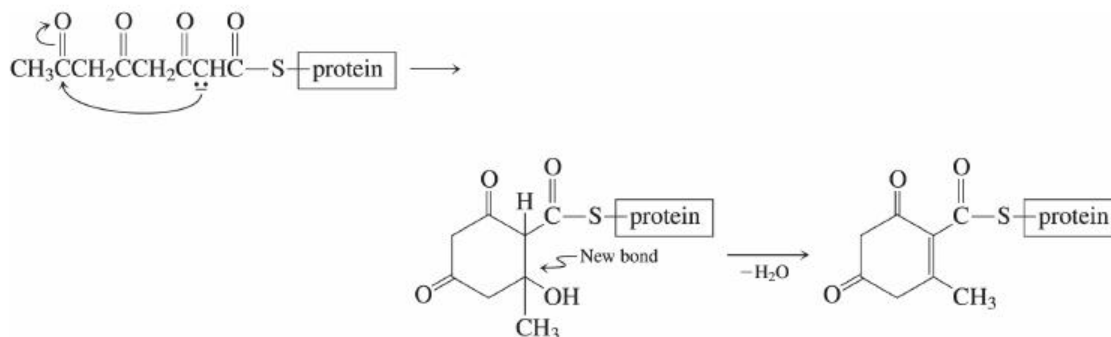
The top mechanism is an $\text{S}_{\text{N}}1$ on the alcohol; the bottom mechanism is the “standard” addition–elimination on the carboxyl group. Labeling the **alcohol** with ^{18}O will distinguish them. The ^{18}O will be lost in the top mechanism, but it will be retained in the bottom one.



57. Start by identifying the carbon atoms in the product that correspond to those in the starting material. Each has a methyl group (*a*) and a carboxy carbon (*b*), so use these as points of reference.



In the structure of the product (above right), note that the carbons *c* and *d* must be the ones that became connected in a ring-forming reaction. They correspond to a methylene and a ketone in the starting material. Because the methylene is an α -carbon with respect to another ketone carbonyl, we know that an aldol condensation can form the bond we need.



All that remains to generate the benzene ring is to enolize both ketones. This step is favored by aromatic stabilization. Finally, hydrolysis of thioester gives the acid.

58. The first part of the problem is just a slight extension of the question in [Problem 46](#): Stereochemistry has been added by extending the chain of the carboxylic acid from five to six carbons. Again, the kinetic advantage in forming a five- as opposed to a six-membered ring is illustrated. The stereochemical outcome results from the carboxylate oxygen attacking the intermediate bromonium ion in a backside manner (did you make a model?!). The second reaction is a double esterification. First, the hydroxy group of one molecule esterifies the carboxy group of another. Then the intermediate (pictured in the middle) undergoes lactonization. Both processes follow the acid-catalyzed mechanism depicted in text Section 19-9.

59. (a)

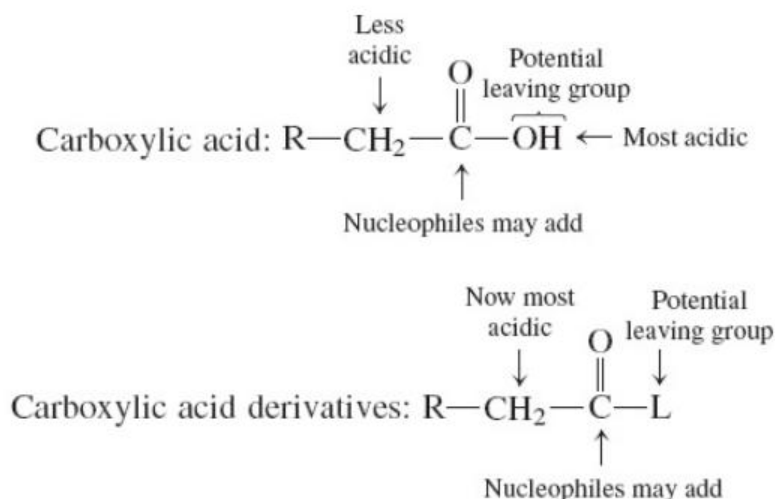
50. (d)

31. (c)

20

Carboxylic Acid Derivatives

Derivatives of carboxylic acids share two properties in common with carboxylic acids themselves: (1) They have a carbonyl group that is susceptible to *nucleophilic addition*, and (2) they have a potential *leaving group* attached to the carbonyl carbon. In addition, they may possess *acidic hydrogens* α to the carbonyl group. However, they **lack** an acidic —OH—OH group on the carbonyl carbon itself. Therefore, it is the α -hydrogens in most carboxylic acid derivatives that are the most readily removed by strong bases.



L = halide (acyl halide)

L = $\overset{\text{O}}{\parallel} \text{OCR}'$ (anhydride)

$\text{L}=\text{OR}'\text{L} = \text{OR}'$ (ester)

$\text{L}=\text{NR}_2'\text{L} = \text{NR}_2'$ (amide)

The mechanisms associated with reactions of these compounds have already been introduced ([Section 19-7](#)), so this chapter mainly presents additional examples of reactions that follow familiar patterns. The most important new points involve (1) the relative reactivities of these derivatives and (2) how to convert each kind of derivative into any of the others, for synthetic purposes.

Outline of the Chapter

[20-1 Relative Reactivities, Structures, and Spectra of Carboxylic Acid Derivatives](#)

Comparisons of physical properties.

[20-2 Acyl Halides](#)

[20-3 Carboxylic Anhydrides](#)

[20-4, 20-5 Esters](#)

[20-6, 20-7 Amides](#)

[20-8 Alkanenitriles](#)

Details of the chemistry of the four major carboxylic acid derivatives, plus nitriles, which are relatives of amides.

Keys to the Chapter

20-1. Relative Reactivities, Structures, and Spectra of Carboxylic Acid Derivatives

In this section, differences in the physical properties of the four main carboxylic acid derivatives are highlighted. The principal factor determining these differences is the electronegativity of the atom attached to the carbonyl carbon. These differences are reflected in a consistent way throughout all the reactions of these compounds: deprotonation of the α -carbon, ~~α -carbon~~, addition to the carbonyl carbon, and protonation of the carbonyl oxygen. Make sure you understand the conceptual arguments here before moving ahead to the specific sections on each derivative. Then, refer back to this text section from time to time as you go through the chapter to see how the reaction details presented in the upcoming sections reflect the general principles outlined here.

20-2. Acyl Halides

Acyl halides are very useful in synthesis because (1) they are easily made from carboxylic acids ([Section 19-8](#)) and (2) they are easily turned into every major type of carbonyl compound, including aldehydes, ketones, and all the other carboxylic acid derivatives. Of all the carboxylic acid derivatives, acyl halides possess the most reactive carbonyl group toward nucleophilic addition and the best leaving group (halide).

20-3. Carboxylic Anhydrides

Carboxylic acid anhydrides are used in synthesis less often than are acyl halides. They are generally not as easily prepared, and they are less reactive. A major drawback to using an anhydride in synthesis is that only one of its carbonyl groups can add a nucleophile; the other falls off as part of the carboxylate ion leaving group (see “Typical Reactions of

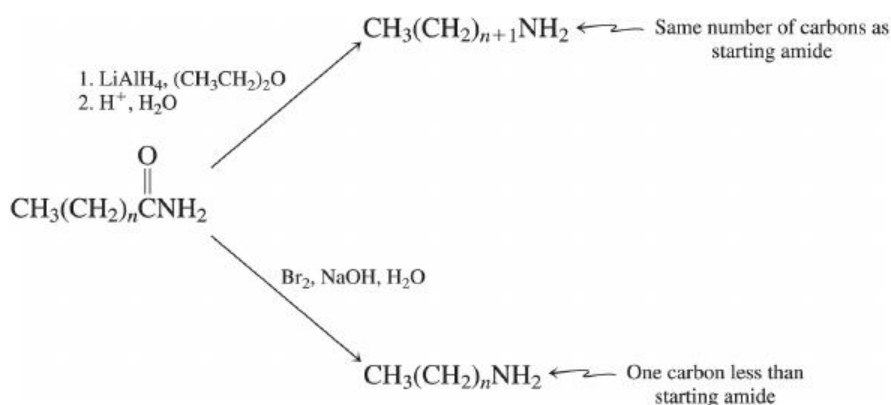
Anhydrides”). Cyclic anhydrides are more useful because the two ends still stay stuck together by a carbon chain after an addition takes place (see “Nucleophilic Ring Opening of Cyclic Anhydrides”).

20-4 and 20-5. Esters

Esters are the most prevalent carboxylic acid derivatives in nature. They also are convenient compounds for synthetic purposes because their intermediate reactivity makes them easy to prepare and store for later use. Acyl halides, in contrast, are so reactive toward water that it takes some care to prevent them from hydrolyzing to some extent upon extended storage. Esters are easily prepared and readily converted into many types of compounds by reactions at both their carbonyl and their α -carbons ([Chapter 23](#)).

20-6 and 20-7. Amides

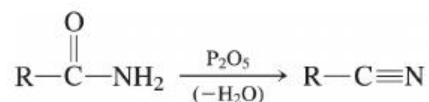
Amides are much less reactive than esters toward nucleophilic acyl substitution. They have greater resonance stabilization by the nitrogen lone pair, and they have a poorer leaving group (NH_3^+ in acid, and NH_2^- in base). Amides do show some special reactions: the possibility of deprotonation of the nitrogen in 1° or 2° amides, reduction reactions that may form either amines or aldehydes, and a new process, the Hofmann rearrangement, which forms an amine with one carbon less than the original amide. Consider the following two amine syntheses.



This Hofmann rearrangement has a mechanism that is related conceptually to some you've seen before but has some very new and unusual details. It merits a careful look, especially the final steps.

20-8. Alkanenitriles

The nitrile functional group is, formally, a dehydrated primary amide. In fact, a little-used nitrile synthesis involves amide dehydration:



In this sense, therefore, it is related to carboxylic acids and their derivatives. The importance of nitriles in synthesis derives from the fact that the nitrile carbon may be introduced into a molecule as the *nucleophilic* cyanide ion in an $\text{S}_{\text{N}}2$ reaction. Then, by the reactions in this text section, the nitrile may be converted to carboxylic acid derivatives, ketones, or aldehydes, in which the originally nucleophilic cyanide carbon has become an *electrophilic* carbonyl carbon. This is a good way to use a nucleophilic substitution to introduce a carbonyl carbon into a molecule.

